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Testing the Bus–DeMeo Asteroid Taxonomy Using Meteorite Spectra

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Abstract

The most widely used method to spectrally classify asteroids is the Bus–DeMeo taxonomy. To test how well the Bus–DeMeo taxonomy groups asteroids on the basis of their mineralogy, we have classified ∼1500 meteorite spectra using this Bus–DeMeo system. Some asteroid classes group together meteorites with similar compositions better than others. Howardite, eucrite, and diogenite spectra tend to be classified as V-types, while ordinary chondrite spectra tend to be classified as S-complex or Q-type bodies. The relatively featureless D- and X-types tend to be dominated by CM carbonaceous chondrites but with a substantial number of matches also with iron meteorites. The large proportion of CM chondrite matches for the D- and X-classes is most likely due to the large number of CM chondrite spectra and the rarity of spectra of more fragile carbonaceous chondrites in our data set. A number of relatively featureless asteroid classes like the C-, B-, L-, and Xc-types group meteorite types together with a wide variety of mineralogies and thermal histories. Visual albedos are vital for distinguishing between many of these assemblages. The Bus–DeMeo taxonomy does have trouble classifying olivine-dominated meteorites that do not have red-sloped spectra because this type of spectrum is rare among asteroids. For many asteroid classes, care must be used when making mineralogical interpretations based solely on spectral type.

Unified Astronomy Thesaurus concepts: Asteroid belt (70); Meteorites (1038)

Materials only available in the online version of record: machine-readable table

1. Introduction

Asteroids are typically classified today by the shape of their reflectance spectra in the visible and near-infrared wavelength regions. The most widely used classification system today is the Bus–DeMeo taxonomy (DeMeo et al. 2009), which was developed using 371 asteroid reflectance spectra. The Bus–DeMeo taxonomy is an extension of the Bus & Binzel (2002) taxonomy into the near-infrared wavelength region, which was itself a modification of the Tholen (1984) taxonomy for charge-coupled device spectra. The Bus–DeMeo taxonomy uses principal component analysis (PCA) and the presence or absence of specific absorption bands to classify a spectrum. PCA is a statistical method for transforming a large number of variables into a much smaller number of principal components, which are linear combinations of the original variables, while retaining most of the information of the original data set. Although new taxonomies have been proposed (e.g., Mahlke et al. 2022; Dyar et al. 2023), the Bus–DeMeo taxonomy is currently the most popular way of classifying an asteroid reflectance spectrum in the visible and near-infrared wavelength region. The Bus–DeMeo taxonomy is extensively used because the classification uses both the visible and near-infrared segments of a reflectance spectrum and can be easily implemented using a web-based classification tool.

The Bus–DeMeo taxonomy broadly separates asteroids into three complexes (S-complex, C-complex, and X-complex) plus nine end-member classes (A, D, K, L, O, Q, R, T, and V; Figure 1). Each complex is broken down into a number of subtypes based on increasingly more subtle spectral differences. There are now 25 different asteroid classes, including the newly designated Xn-class (Binzel et al. 2019). The letter *w* is added to the class for extremely red-sloped spectra, where red refers to the reflectance tending to increase with increasing wavelength. A spectrum is classified as indeterminate (U) when it cannot be placed in a Bus–DeMeo class. Visual albedo is not used in the Bus–DeMeo taxonomy.

Since the Bus–DeMeo taxonomy classifies asteroids into groups according to their spectral properties, the apparent correlation between the taxonomic classes and the asteroids' mineralogy has motivated some researchers to use the classifications to estimate asteroids' compositions, even though this correlation is not 100% definitive. DeMeo et al. (2022) looked at spectral similarities between 500 asteroid spectra and over 1000 specific meteorite spectra and used chi-square fitting to find spectral matches with high statistical significance between asteroid and meteorite spectra. In contrast, in this paper, we do the reverse, classifying meteorite spectra using the Bus–DeMeo taxonomy. Both studies seek to determine how well each asteroid class groups objects mineralogically. The primary difference between the two studies is that DeMeo et al. (2022) looked for the best spectral matches to members of an asteroid class, while our study focuses on characterizing the potential mineralogic diversity with an asteroid class.

We use a data set of meteorite spectra from the Reflectance Experiment Laboratory (RELAB) database (Milliken 2020) plus a small number of RELAB spectra recently collected by Dyar et al. (2023). The spectra range from those with strong absorption bands (e.g., eucrites, ordinary chondrites) to those with much weaker to absent features (e.g., CM chondrites,

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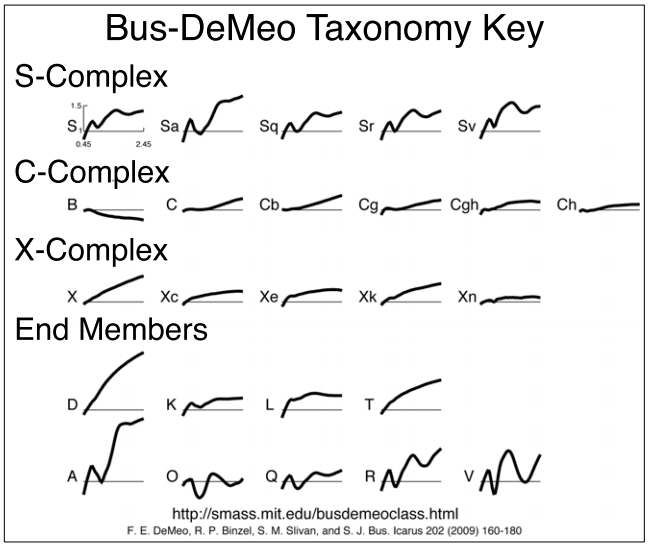


Figure 1. The Bus–DeMeo taxonomy key displays reflectance spectra for each of the taxonomic classes. The wavelength range of each spectrum is 0.45–2.45 μm. The figure is used with the permission of Francesca DeMeo.

irons). High-resolution spectral measurements like those analyzed in DeMeo et al. (2022) often allow for detailed mineralogical interpretations. However, our study hopes to constrain the mineralogical range for asteroids that are classified using lower-resolution photometric measurements (e.g., Choi et al. 2023). What does a taxonomic designation really tell us about the mineralogy of an asteroid?

Very few asteroids have compositions that are definitively known, making it difficult to test classifications and their accuracies. Samples have been returned for only three near-Earth asteroids (NEAs). LL chondrite material (Nakamura et al. 2011) was returned from S-type (25143) Itokawa by the Hayabusa mission. CI chondrite material (Yada et al. 2022; Yokoyama et al. 2022; Greenwood et al. 2023) was returned from C-complex (162173) Ryugu (Sugita et al. 2019; Tatsumi et al. 2021) by the Hayabusa2 mission. Type 1 carbonaceous chondritic material (e.g., King et al. 2024) from B-type (101955) Bennu (Clark et al. 2011) was collected by the OSIRIS-REx mission and returned back to Earth. Asteroidal fragments were also recovered from NEA 2008 TC_3 that exploded in the Earth's atmosphere over Sudan (Jenniskens et al. 2009). That meteorite, called Almahata Sitta, was classified as an anomalous ureilite, but it also contained a significant amount of chondritic material (Bischoff et al. 2010). Before the encounter with Earth, 2008 TC_3 was spectrally characterized in the visible (Jenniskens et al. 2009) and classified as an F-type from the Tholen (1984) taxonomy. Today, F-types are usually classified as B-types (Alí-Lagoa et al. 2013).

In this paper, we seek to relate the known mineralogy of meteorites to the classifications of the Bus–DeMeo taxonomy by applying the latter to 1489 RELAB meteorite reflectance spectra. The goal of this work is to find correlations between different meteorite groups and asteroid classes to see how well each asteroid class is able to group meteorites with similar mineralogies based on their reflectance spectra. We treat the meteorite spectral data set just like an asteroid spectral data set where we do not know the mineralogies of the objects and, therefore, just classify the spectra according to their spectral properties. But since we do know the meteorite classification for each spectrum, we can use that information to give insight on how well the Bus–DeMeo taxonomy mineralogically classifies reflectance spectra.

2. Data

Meteorite reflectance spectra (including spectra from lunar and Martian meteorites) from the RELAB spectral database (Milliken 2020) were downloaded. We also used some newly acquired RELAB meteorite spectra that were part of the Dyar et al. (2023) machine-learning project to classify asteroid spectra using meteorite spectra as the training set. Meteorite classifications were taken from the Meteoritical Bulletin Database (2024) except where meteorites (EET 87532, QUE 97002) were reclassified (Righter 2022).

Spectra of lunar meteorites were included based upon the recent observation that NEA (469219) Kamo'oalewa has a reflectance spectrum (Sharkey et al. 2021) and orbit (Castro-Cisneros et al. 2023) consistent with a lunar origin. Asteroids derived from Mars also appear plausible (Polishook et al. 2017; Hyodo & Genda 2018). The Bus–DeMeo taxonomy was defined just using asteroid spectra, so it is expected that the taxonomy may have difficulty in classifying lunar and Martian meteorite spectra.

Our goal was to use meteorite reflectance spectra from samples that could potentially represent the surface of an asteroid. Spectra that were labeled as terrestrially weathered or rusted were not used. Spectra with extremely large uncertainties were also not used. Reflectance spectra were included for both powders and chips/slabs because so little is known about the range of surface properties of asteroids. The samples (21%) measured as chips or slabs were often problematic because of difficulties in powdering some iron-dominated meteorites and/or the rarity of those samples. A few (5%) of the meteorites containing metallic iron had spectra measured of just the silicate portion. About 1% of the samples were acid-washed to try to remove terrestrial weathering products. We included 3% of the samples that were heated to higher temperatures, primarily to remove adsorbed water. Roughly 3% of the meteorites used had been irradiated with a laser to simulate space weathering. Some repeat spectra were also included of samples that were remeasured. A total of 1498 meteorite spectra for 654 different meteorites were part of this study. The meteorite spectra used in this analysis are listed in Table A1 in the Appendix.

Table 1 shows the distribution of meteorite spectra used in this study. The number of spectra in the RELAB database for any meteorite type is a function of many factors, such as scientific interest, availability, and ease of preparation. These biases will affect all percentages calculated in this paper; however, it is extremely difficult to debias the results, since even the fall statistics will be biased due to fragile material preferentially not reaching the Earth's surface due to the Earth's atmosphere. Meteorite classes that appear to come from the same parent body and/or are compositionally and isotopically similar, such as the howardites, eucrites, and diogenites (HEDs) (Mittlefehldt 1997); acapulcoites/lodranites (McCoy et al. 1997); and CB/CH chondrites (Rüfenacht et al. 2023), are grouped together. The C1/C2-ungrouped meteorites include C1/2-ungrouped and C2-ungrouped meteorites. The C3/C4/C5-ungrouped meteorites include C3-ungrouped, C4-ungrouped, and C4/5 meteorites. Meteorites classified as

Table 1
Number and Percentage of Each Meteorite Group Found among the Meteorite Spectra Used in the Analysis

Meteorite Group	Number of Spectra	%	Fall %
HED	265	17.8	5.9
L	206	13.8	38.7
CM	171	11.5	1.8
H	132	8.9	33.7
LL	99	6.6	8.3
CV	82	5.5	0.6
Ureilite	67	4.5	0.5
Martian	59	4.0	0.4
Acapulcoite/lodranite	53	3.6	0.2
Lunar	53	3.6	0.0
C1/C2-ungrouped	33	2.2	0.5
EH	29	1.9	0.8
Aubrite	27	1.8	1.0
CO	25	1.7	0.6
CR	25	1.7	0.3
CK	24	1.6	0.2
EL	23	1.5	0.7
Iron	23	1.5	4.1
CI	19	1.3	0.4
R	18	1.2	0.1
Pallasite	14	0.9	0.3
Mesosiderite	13	0.9	0.6
CB/CH	10	0.7	0.1
Brachinite	7	0.5	0.0
Angrite	4	0.3	0.1
Winonaite	3	0.2	0.1
C3/C4/C5-ungrouped	2	0.1	0.1
K	2	0.1	0.1
Achondrite-ungrouped	1	0.1	0.1

Note. The groups are ordered from highest to lowest number. The fall percentage for each meteorite group is also included.

impact melts are included with the meteorite type that was the precursor.

The large percentage of HED spectra in our study is related to background studies for the Dawn spacecraft mission to (4) Vesta (e.g., Reddy et al. 2012), which has long been known to be spectrally similar to the HEDs (e.g., McCord et al. 1970; Larson & Fink 1975; Hiroi et al. 1994). Ordinary chondrite meteorites (H, L, LL) are also well represented since they make up ∼80% of all falls. CM chondrites also tend to be well represented, which is probably partially due to the preparation for sample-return missions (Hayabusa2, OSIRIS-REx) to two C-complex bodies. The relatively large percentage of ureilites is due to the many reflectance measurements of Almahata Sitta fragments. As expected, meteorites that rarely fall to Earth are also rare in the RELAB spectral database. Meteorites that contain large abundances of metallic iron and are hard to powder are also rarer in the RELAB spectral database.

3. Analysis

The meteorite reflectance spectra were primarily classified using a UNIX-based version of the Bus–DeMeo web-based classifier, which can classify multiple spectra. The inputted spectral files contained either two rows of data (wavelength in nanometers and reflectance for each wavelength) or three rows of data (wavelength in nanometers, reflectance for each wavelength, and uncertainty for each reflectance). Some of the older RELAB spectra did not have available reflectance uncertainties. The classification program would use the reflectance uncertainties in classifying the spectrum when those data were available. The resulting classification for a particular reflectance spectrum could change depending on whether the reflectance uncertainties were included with the inputted spectrum. These classification changes due to using or not using the reflectance uncertainties tended to be between different classes in the same complex (S-, C-, or X-).

The Bus–DeMeo classification program first smooths each spectrum and then samples each spectrum to produce 41 data points from 0.45 to 2.45 μm at 0.05 increments (DeMeo et al. 2009). The spectra are normalized to unity at 0.55 μm. A fitted slope is then divided out of each spectrum. Each spectrum is classified using 40 data points with the removal of the 0.55 μm reflectance value, which will be unity for all spectra. A covariance matrix that was originally determined by the original asteroid classification in DeMeo et al. (2009) is used to calculate principal components for each spectrum. Each spectrum is classified using the spectrum's fitted slope and five principal components.

Spectra that were not uniquely classified are assigned a set of possible classifications. For those cases, it was necessary to run them individually through the web-based program, followed by visual classification. Approximately 40% of the meteorite spectra had to be visually classified. The visual classification of spectra contributes considerable uncertainty to the process because different people may visually classify any given spectrum another way even with given guidelines. Some of the choices were obvious (e.g., A- or D-type), while others were much more subtle (e.g., Xk-, Xc-, Xe-, C-, Ch-, or Xn-type; L- or Xe-type). Most spectra were visually classified by at least two people, and a discussion was held to produce a consensus classification if needed. In some cases, none of the given choices for a meteorite spectrum (e.g., Xk-, Xc-, Xe-, C-, Ch-, or Xn-type; L- or Xe-type) adequately reflected its spectral characteristics. For those "anomalous" meteorite spectra, the best match to an asteroid class was selected even if the spectral properties of the class did not correspond "well" to the spectrum of the meteorite because the visual classification process does not currently allow for the choosing of a class (e.g., U) that is outside the given choices.

4. Results

4.1. Overview

The numbers and percentages of meteorite spectra for each classification are listed in Table 2, along with the percentages for each class found in the DeMeo et al. (2009) and Binzel et al. (2019) studies. For percentages derived from Binzel et al. (2019), we only included bodies that had a similar spectral coverage and were given a definite Bus–DeMeo classification. DeMeo et al. (2009) classified primarily main-belt objects, while Binzel et al. (2019) classified Mars-crossers and NEAs. Comparing percentages (excluding the U, Xn, and all classes with a w designation), the distribution found among the meteorite matches appears more similar to the NEA/Mars-crosser (Binzel et al. 2019) distribution than the main-belt distribution of asteroid classes. The classified NEA/Mars-crosser population is much smaller in diameter than the main-belt asteroid population and has orbits more similar to the Earth's, which may explain why the distribution of asteroid

Table 2
Numbers and Percentages of Asteroid Classes that Were Determined from the Meteorite Spectra

Class	Number	%	DeMeo et al. (2009) %	Binzel et al. (2019) %
V	267	17.9	4.6	5.0
Q	234	15.7	2.2	13.5
K	191	12.8	4.3	0.3
B	107	7.2	1.1	3.0
Xc	98	6.6	0.8	0.3
Sq	97	6.5	7.8	11.9
S	83	5.6	38.8	18.8
L	56	3.8	5.9	4.0
X	51	3.4	1.1	4.0
Sr	48	3.2	5.9	7.6
C	41	2.8	3.5	4.0
D	38	2.6	4.3	2.0
Xk	31	2.1	4.9	1.3
U	27	1.8	⋯	1.3
R	22	1.5	0.3	0.0
Xe	19	1.3	1.9	2.0
Cgh	11	0.7	2.7	0.0
Cg	9	0.6	0.3	0.3
Cb	7	0.5	0.8	0.3
Sa	7	0.5	0.5	0.7
Svw	7	0.5	⋯	0.7
Vw	7	0.5	⋯	0.0
Sv	6	0.4	0.5	0.7
O	5	0.3	0.3	0.0
A	4	0.3	1.6	0.3
Ch	4	0.3	4.9	0.7
Qw	3	0.2	⋯	0.3
Sw	3	0.2	⋯	10.2
Xn	3	0.2	⋯	0.3
T	2	0.1	1.1	0.0
Srw	1	0.1	⋯	2.3

Note. The classes are ordered from highest to lowest number of meteorites with that classification. Also included are the percentages for each class from the DeMeo et al. (2009) study of primarily main-belt asteroids and the Binzel et al. (2019) study of near-Earth asteroids and Mars-crossers.

classes derived from the meteorite spectra tend to be more similar to the distribution found among NEAs/Mars-crossers than main-belt asteroids.

The large number of V-type matches is due to the large number of HEDs in the sample. Both V-types and HEDs have strong low-Ca pyroxene absorptions. The large number of Q-type matches results from the large number of ordinary chondrites in the sample because both Q-types and some ordinary chondrites have relatively strong absorptions due to olivine and low-Ca pyroxene. The large number of K-type matches arises because both K-types and many meteorite types typically have relatively flat spectra with subdued bands due to olivine and low-Ca pyroxene.

The small number of matches for some asteroid classes is also consistent with what is known about those classes. Matches with A-types are rare due to the rarity of olivine-dominated meteorite spectra with red spectral slopes. Matches with Ch- and Cgh-types are also rare. The letter *h* in the classification refers to the presence of a 0.7 μm band. CM2 chondrites often have this 0.7 μm band (e.g., Cloutis et al. 2011), but the strength of this feature can vary. Xn-types like (44) Nysa that have a sharp 0.9 μm feature (Figure 1) are rare among meteorite spectra. O-types are known to have very distinctive near-infrared spectral properties (e.g., DeMeo et al. 2009; Burbine et al. 2011). Matches with red-sloped T-types are also rare among meteorite spectra. This paucity of matches with T-types may be due to the failure of this type of material, which is expected to be organic-rich and fragile, to make it through Earth's atmosphere.

The percentages of matches for each asteroid class for members of a particular meteorite group are listed in Table 3. For example, 88% of the meteorite spectra classified as V-types are from HEDs, while 8% of the meteorite spectra classified as V-types are from Martian meteorites. Only classes with $\geqslant$38 matches are listed, so the possible inclusion of an "anomalous" meteorite spectrum with "unusual" spectral properties does not dramatically affect the percentages. For each asteroid class, we also arbitrarily list only meteorite matches of 5% or greater. The percentages of matches for each asteroid class are a function of the number of meteorite spectra used in the analysis and the type of meteorite spectra included. For example, the spectral library used here has a relatively large percentage ($\sim$7%) of lunar and Martian meteoritic material, which would be expected to be rare among asteroids. The spectra of these lunar and Martian meteorites range from having quite distinctive mineral absorption bands to being extremely space-weathered with very subdued features.

Larger percentages of matches for any particular asteroid class give more confidence that the class is likely composed of that particular mineralogy. It must be noted that despite the apparent exactness of the percentages in Table 3, these results are very qualitative. The calculated percentages are a function of the distribution of different meteorite types in this sample suite of meteorite spectra, and the percentages will change if different spectra are part of the analyzed data set. The need to visually determine the classifications of many of the spectra also adds uncertainty in the calculated percentages because different people may visually classify the same spectrum differently.

4.2. Individual Class Characteristics

Each of these asteroid classes and the meteorite spectra assigned those classifications are discussed below. We present the results in order according to those with the highest percentage of matches for a particular meteorite group to those with the lowest percentage. For each of these taxonomic classes, a typical asteroid spectrum is shown along with a few of the meteorite spectral matches. Reflectance error bars for the meteorites are not displayed since they tend to be relatively small. The asteroid reflectance spectrum is multiplied by the visual geometric albedo to facilitate plotting on a similar scale to the unnormalized meteorite spectra and to show where visual albedo may be important in discriminating between different mineralogies.

Visual albedo is important for the mineralogical characterization of asteroids, even though it is not part of the Bus–DeMeo classification process. For example, opaque-rich meteorites (e.g., carbonaceous chondrites) tend to have lower visual albedos than silicate-rich meteorites (e.g., HEDs, ordinary chondrites, aubrites; Ostrowski & Bryson 2019). Table 4 lists the mean Near-Earth Object Wide-field Infrared Survey Explorer (NEOWISE) visual geometric albedos for the asteroids classified using the Bus–DeMeo taxonomy. Table 4 shows that S-complex bodies tend to have higher visual albedos than X-complex bodies, and X-complex bodies tend to have higher visual albedos than C-complex bodies.

Table 3
Percentages of Matches for Each Asteroid Class for Members of a Particular Meteorite Group

Bus–DeMeo Class	Percentage of Each Meteorite Type	Best Spectral Analog
V	HED (88%); Martian (8%)	HED
D	CM (50%); iron (39%); CO (5%)	CM or iron
X	CM (45%); iron (16%); C1/C2-ungrouped (10%); EH (8%); pallasite (8%)	CC or iron-dominated
Sq	L (42%); H (29%); LL (16%); ureilite (6%)	OC
Sr	L (38%); H (25%); HED (17%); aca/lod (13%); Martian (6%)	L or H or HED
S	L (36%); H (33%); LL (16%); aca/lod (8%)	OC
Q	L (30%); LL (23%); H (14%); ureilite (11%); lunar (8%); Martian (6%); HED (5%)	OC
K	CM (28%); CV (24%); CK (9%); ureilite (7%); CO (7%); R (6%); lunar (5%)	CC
C	CM (24%); brachinite (12%); L (12%); C1/C2-ungrouped (10%); ureilite (10%); H (7%); LL (5%)	Various
B	CV (18%); aubrite (15%); CM (14%); lunar (13%); L (10%)	Various
L	EL (16%); CI (13%); EH (13%); CB/CH (11%); H (9%); CM (7%); C1/C2-ungrouped (5%); CO (5%); CR (5%); ureilite (5%)	Various
Xc	EH (15%); CM (12%); L (12%); aubrite (10%); H (10%); CV (9%); EL (8%); C1/C2-ungrouped (8%); CI (5%)	Various

Note. Only classes with 38 or more matches are listed. The classes are ordered according to those with the highest percentage match for a particular meteorite or a group of meteorites. We only list meteorite groups with match percentages of 5% or more. From this analysis, we also list what we believe to be the best spectral analog for each asteroid class. Acronyms used include HED for howardites, eucrites, and diogenites; OC for ordinary chondrites (H, L, and LL); CC for carbonaceous chondrites; and aca/lod for acapulcoites/lodranites.

Table 4
The Mean NEOWISE Visual Geometric Albedos for Bus–DeMeo Asteroid Classes from Mainzer et al. (2011)

Class	Objects	Albedo	Standard Deviation
A	5	0.191	±0.034
B	2	0.120	±0.022
C-complex	32	0.058	±0.028
C	9	0.050	±0.006
Cb	1	0.043	…
Cg	1	0.063	…
Cgh	8	0.065	±0.048
Ch	13	0.058	±0.009
D	13	0.048	±0.025
K	11	0.130	±0.058
L	19	0.149	±0.066
O	1	0.339	…
Q	1	0.227	…
R	1	0.148	…
S-complex	121	0.223	±0.073
S	66	0.211	±0.068
Sa	1	0.367	…
Sq	6	0.243	±0.039
Sqw	7	0.231	±0.043
Sr	10	0.266	±0.055
Srw	2	0.279	±0.051
Sv	1	0.309	…
Sw	28	0.221	±0.094
T	2	0.042	±0.004
V	8	0.362	±0.100
X-complex	17	0.111	±0.143
X	3	0.047	±0.060
Xc	2	0.129	±0.077
Xe	4	0.136	±0.238
Xk	8	0.095	±0.038

Note. The table lists the Bus–DeMeo taxonomic class, the number of objects in the Mainzer et al. (2011) sample, the visual geometric albedo, and the standard deviation for the visual geometric albedo.

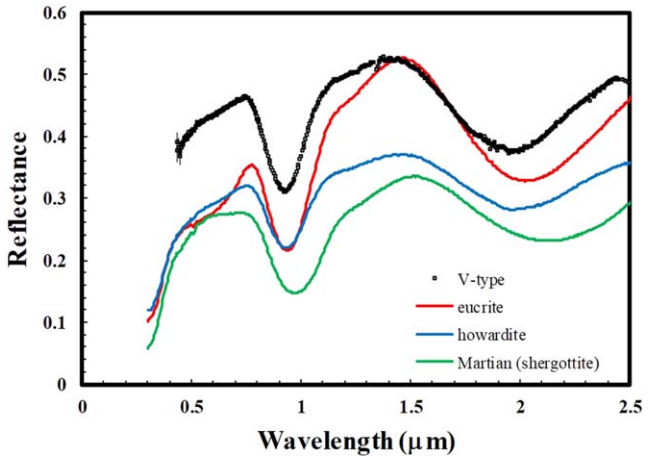


Figure 2. Reflectance spectra of objects classified as V-types. Plotted are (4) Vesta (black open squares; DeMeo et al. 2009), the eucrite Bouvante (<25 μm; red line), howardite QUE 97002 (<75 μm; blue line), and Martian meteorite (shergottite) Zagami (<125 μm; green line). The Vesta spectrum was normalized to unity at 0.55 μm and then multiplied by 0.4228 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

4.2.1. V-types

V-type asteroids are linked with HED meteorites due to their distinctive low-Ca pyroxene absorption features (Figure 2). The most-studied V-type asteroid is Vesta, which has long been thought to be the parent body of the HEDs (e.g., McCord et al. 1970; Larson & Fink 1975; McSween et al. 2013). Inner-belt V-types (e.g., Binzel & Xu 1993; Moskovitz et al. 2010), which are typically called Vestoids, are thought to be fragments of Vesta.

Approximately 90% of V-type matches are HEDs (Table 3), consistent with the prevailing opinion about the composition of V-types. Relatively few meteorite groups have members with strong absorption bands due almost entirely to low-Ca pyroxene. Matches with Martian meteorites are also consistent with the low-Ca pyroxene-rich nature of some of these samples. However, Figure 2 shows that spectra of Martian meteorites tend to have much broader pyroxene absorptions than those of Vesta and HEDs.

4.2.2. D-types

D-type asteroids are typically linked with organic-rich carbonaceous chondritic material due to their red spectral

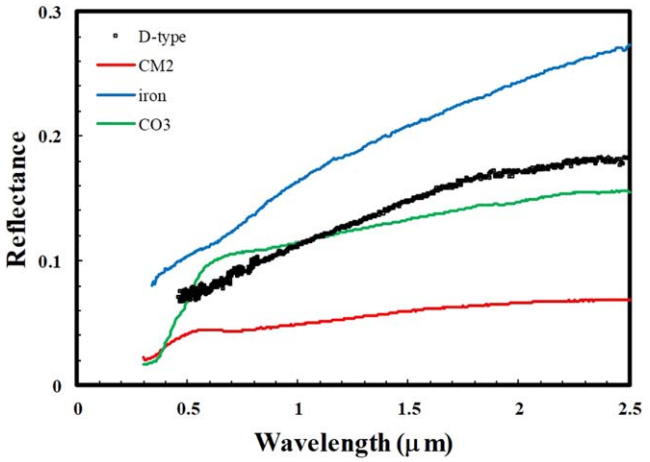


Figure 3. Reflectance spectra of objects classified as D-types. Plotted are (1143) Odysseus (black open squares; DeMeo et al. 2009), CM2 chondrite Mighei (<100 μm; red line), iron meteorite Odessa (45–90 μm; blue line), and CO3 chondrite MIL 07687 (<35 μm; green line). The Odysseus spectrum was normalized to unity at 0.55 μm and then multiplied by 0.0753 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

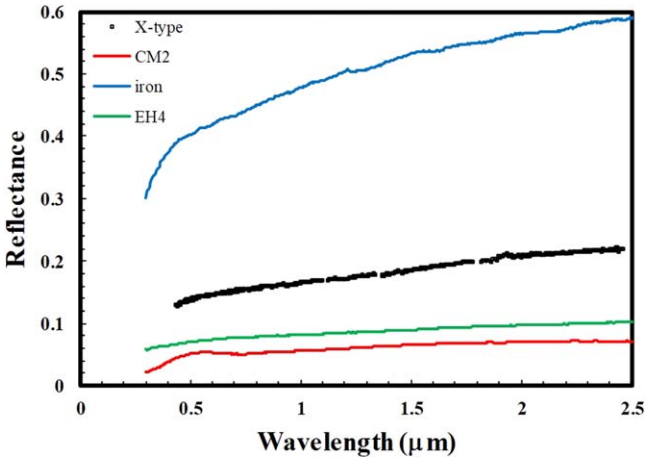


Figure 4. Reflectance spectra of objects classified as X-types. Plotted are (22) Kalliope (black open squares; DeMeo et al. 2009), CM2 chondrite Boriskino (<35 μm; red line), iron meteorite Mundrabilla (flat surface; blue line), and EH4 chondrite Abee (<125 μm; green line). The Kalliope spectrum was normalized to unity at 0.55 μm and then multiplied by 0.1419 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

slopes (Figure 3) and high abundances among Jupiter Trojans, which are located at ~5.2 au. D-types also tend to have very low visual albedos (Table 4; Mainzer et al. 2011). The expected fragile nature of this type of material may make it difficult for it to survive during descent through the Earth's atmosphere. Ungrouped C2 chondrites Tagish Lake (Hiroi et al. 2001) and Tarda (Marrocchi et al. 2021) have been linked with D-types, though Hiroi et al. (2022) did not find a "good" spectral match between Tarda and D-types.

Approximately 50% of D-type matches are with CM chondrites (Table 3). CM chondrite spectra can range from extremely red-sloped to blue-sloped (Cloutis et al. 2011), with the CM chondrites with significant red spectral slopes being classified as D-types. The high percentage of matches with CM chondrites is most likely due to the large number of CM chondrites and the rarity of more fragile organic-rich carbonaceous chondritic material in our data set. However, approximately one-third of the matches are with iron meteorites (Table 3) because irons can also have extremely red-sloped spectra. A number of these iron meteorite matches were for iron slabs, which may not be representative of an iron asteroid's surface. DeMeo et al. (2022) previously saw the spectral similarity between D-types, carbonaceous chondrites, and iron meteorites but noted how iron meteorites would tend to have higher visual albedos (~0.3; Ostrowski & Bryson 2019) than both D-types (0.05; Table 4) and CM and CI carbonaceous chondrites (~0.06).

4.2.3. X-types

X-types are typically linked with iron meteorites because both types of objects have slightly red-sloped reflectance spectra without distinctive absorption features (Figure 4). In this study, most of the X-type matches are with carbonaceous chondrites (Table 3) because many carbonaceous chondrite spectra have relatively featureless spectra. As with the D-type matches, the high percentage of matches with CM chondrites is most likely due to the large number of CM chondrites in our data set. However, ~35% of the matches are with meteorites with significant metallic iron abundances such as irons, enstatite chondrites, and pallasites. The large number of carbonaceous chondrites in the spectral data set studied here biases the results toward carbonaceous chondrites. The carbonaceous chondrite matches tend to have lower visual albedos (~5%) compared to meteorites with significant iron components (e.g., irons, enstatite chondrites, and pallasites; Ostrowski & Bryson 2019). The CM2 chondrite spectral matches also tend to have a feature at 0.7 μm (Figure 4); however, Ch or Cgh was not given as an option during visual classification.

4.2.4. Sq-types

Sq-types are typically linked with ordinary chondrites (e.g., DeMeo et al. 2022) because these bodies have spectral characteristics intermediate between S- and Q-types (Figure 5). S-types typically have reflectance spectra consistent with space-weathered ordinary chondrites (e.g., Chapman 1996; Eschrig et al. 2022), while Q-types typically have reflectance spectra similar to unweathered ordinary chondrites (e.g., Binzel et al. 1996). Space-weathered ordinary chondrites have redder spectral slopes and weaker absorption bands than "fresher" ordinary chondrites.

Approximately 90% of Sq matches are with ordinary chondrites (Table 3), which is also consistent with the prevailing notion about Sq bodies. Most of the Sq matches are with L chondrites; however, L chondrites are the largest number of ordinary chondrites in the data set. Because there are 74 more L chondrites than H chondrites, these results are biased toward matches with L chondrites. Approximately 7% of the matches are with ureilites (Figure 5), which also have absorption features due to olivine and low-Ca pyroxene.

4.2.5. Sr-types

Sr-types would be expected to be linked with either ordinary chondrites or primitive achondrites due to having spectral properties intermediate between S- and R-types (Figure 6). As mentioned earlier, S-types are typically linked with ordinary chondrites. R-types have spectral properties consistent with low-Ca pyroxene–olivine assemblages that have extremely

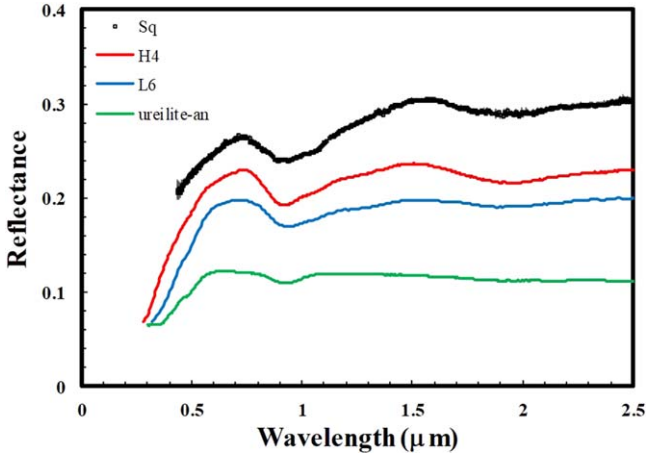


Figure 5. Reflectance spectra of objects classified as Sq-types. Plotted are (3) Juno (black open squares; DeMeo et al. 2009), H4 chondrite Monroe (<125 μm; red line), L6 chondrite Karkh (<150 μm, metal removed; blue line), and the anomalous ureilite MET 01085 (<75 μm; green line). The Juno spectrum was normalized to unity at 0.55 μm and then multiplied by 0.2383 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

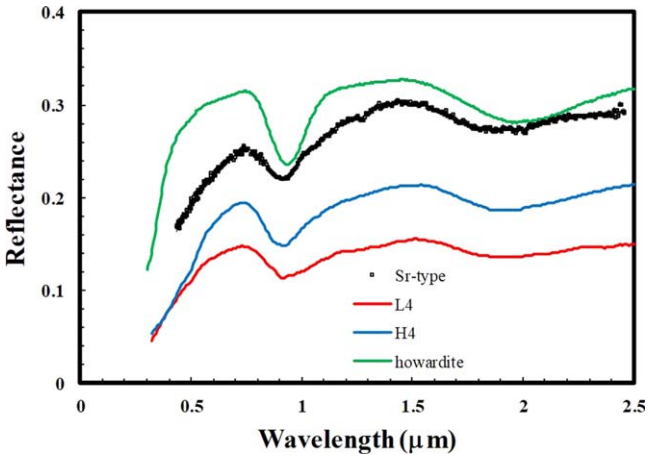


Figure 6. Reflectance spectra of objects classified as Sr-types. Plotted are (237) Coelistina (black open squares; DeMeo et al. 2009), L4 chondrite Rio Negro (<150 μm, metal removed; red line), H4 chondrite Chela (<150 μm, metal removed; blue line), and howardite PCA 02015 (<40 μm; green line). The Coelistina spectrum was normalized to unity at 0.55 μm and then multiplied by 0.2108 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

distinctive pyroxene features. R chondrites have been typically linked with primitive achondrites (Abell 2003).

Over 60% of the Sr matches are with ordinary chondrites (H and L; Table 3), consistent with the prevailing notion about Sr bodies (DeMeo et al. 2022). The remaining matches are with a variety of achondrites such as HEDs and acapulcoites/lodranites.

4.2.6. S-types

As mentioned earlier, S-types are typically linked with ordinary chondrites (Figure 7). Approximately 85% of the S-type matches are with ordinary chondrites (Table 3), consistent with the prevailing notion about S-type bodies having mineralogies similar to ordinary chondrites (e.g., Chapman 1996; Eschrig et al. 2022). Approximately 10% of the matches are with acapulcoites/lodranites.

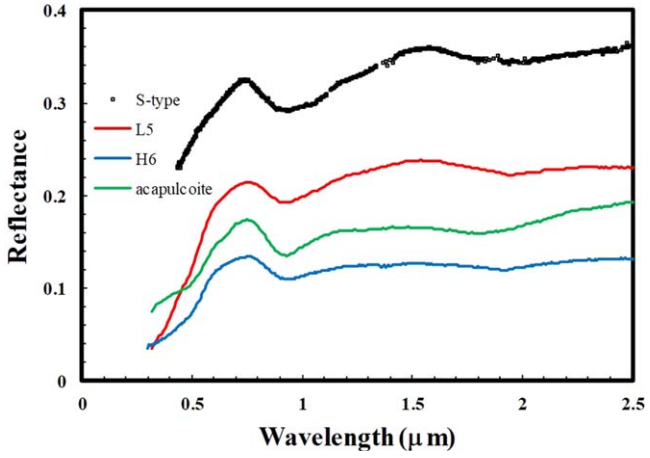


Figure 7. Reflectance spectra of objects classified as S-types. Plotted are (7) Iris (DeMeo et al. 2009; black open squares), L5 chondrite Blackwell (<150 μm; red line), H6 chondrite Ozona (<150 μm; blue line), and acapulcoite Dhofar 125 (<125 μm, treated with ethanolamine thioglycolate to remove terrestrial weathering; green line). The Iris spectrum was normalized to unity at 0.55 μm and then multiplied by 0.2766 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

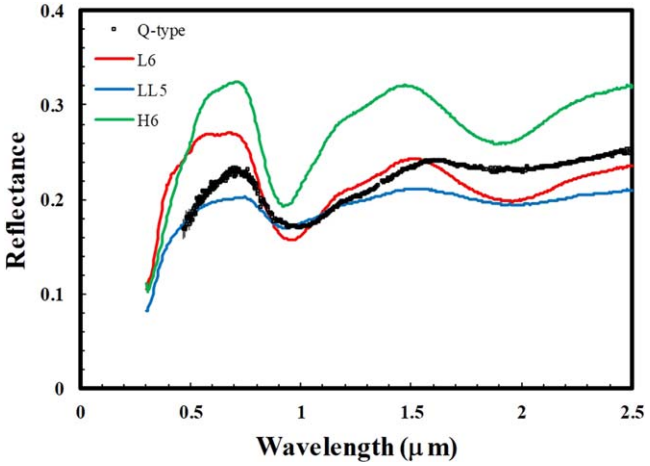


Figure 8. Reflectance spectra of objects classified as Q-types. Plotted are (1862) Apollo (DeMeo et al. 2009; black open squares), L6 chondrite Harleton (125–500 μm; red line), LL5 chondrite Chelyabinsk (<125 μm; blue line), and H6 chondrite Kernouve (<125 μm; green line). The Apollo spectrum was normalized to unity at 0.55 μm and then multiplied by 0.20 (Rozitis et al. 2013). The error bars for the asteroid spectrum are 1σ.

4.2.7. Q-types

As noted above, Q-types have spectral features consistent with ordinary chondrites (Figure 8). Approximately 65% of the Q-type matches are with ordinary chondrites (Table 3), consistent with current thinking that Q-type bodies have mineralogies similar to those of ordinary chondrites (e.g., Binzel et al. 1996). Like the Sq-type matches, the prevailing match is with L chondrites; however, the relatively large number of L chondrites in the database biases these results.

4.2.8. K-types

K-types are typically linked with CO and CV chondrites (Bell 1988; Clark et al. 2009) because both have weak absorption bands due to olivine (Figure 9). Partially

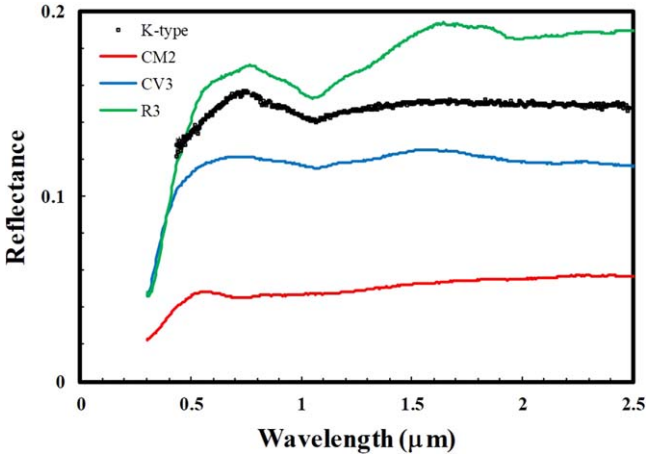


Figure 9. Reflectance spectra of objects classified as K-types. Plotted are (221) Eos (DeMeo et al. 2009; black open squares), CM2 chondrite LEW 87148 (<125 μm; red line), CV3 chondrite ALH 84028 (<125 μm; blue line), and R3 chondrite PRE 95411 (<125 μm; green line). The Eos spectrum was normalized to unity at 0.55 μm and then multiplied by 0.1400 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

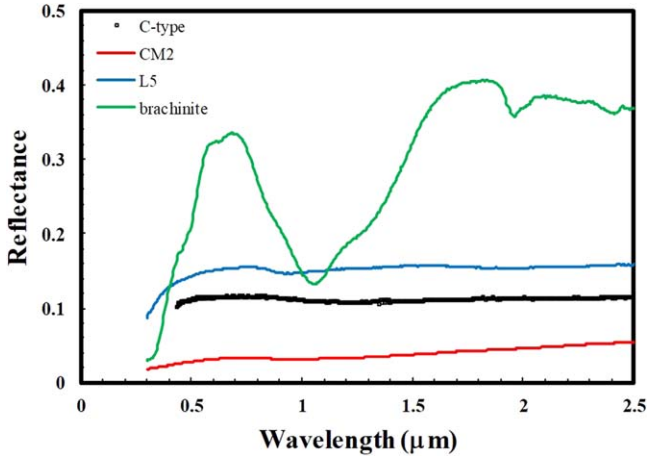


Figure 10. Reflectance spectra of objects classified as C-types. Plotted are (1) Ceres (DeMeo et al. 2009; black open squares), CM2 chondrite Murchison (<63 μm, heated to 500°C; red line), L5 chondrite Tsarev (<40 μm, laser irradiated; blue line), and brachinite Brachina (<63 μm, cleaned with HCl; green line). The Ceres spectrum was normalized to unity at 0.55 μm and then multiplied by 0.1132 (Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ. Even though brachinite is not spectrally similar to C-types, it is classified as a C-type because the other given options (X, Xe, Xk, Xn) are much worse spectral matches.

differentiated material, CK chondrites, and R chondrites have also been proposed as possible analogs for K-types (e.g., Mothé-Diniz & Carvano 2005; Mothé-Diniz et al. 2008; Krämer Ruggiu et al. 2021). Approximately 75% of the K-type matches are with carbonaceous chondrites (Table 3), consistent with the notion that K-type bodies have mineralogies similar to CO, CV, and CK chondrites (Bell 1988; Clark et al. 2009). However, most of the matches are with CM chondrites, which was not expected. The large number of matches with CM chondrites is due to the visual classification only giving K- or Xe-type as an option for many of the CM chondrite spectra. Approximately 6% of the matches are with R chondrites (Table 3).

4.2.9. C-types

C-types are typically linked with carbonaceous chondrites because they have relatively flat and featureless spectra (Figure 10). A variety of different types of meteorites do match with the C-types (Table 3). CM chondrites have the largest number of matches, though a number of meteorite types (e.g., ordinary chondrites, ureilites) that can have weak mafic absorption bands can also be classified as C-types.

Even though brachinites have strong olivine absorption bands, we classify them as C-types because the other options (X, Xe, Xk, Xn) are much worse spectral matches. Brachinite spectra do not typically classify as A- or Sa-types (Figure 1) because their spectra are not red-sloped. The Bus–DeMeo classification system fails in classifying relatively flat olivine-dominated spectra because the fitted slope and five principal components for these spectra are not adequately separating these objects from weakly featured spectra. This classification issue argues for an option to classify a spectrum as U when the spectrum is not a good match for any of the proposed classes.

4.2.10. B-types

B-types are typically linked with type 1 or 2 carbonaceous chondritic material (Clark et al. 2010) due their relatively featureless spectra and blue spectral slopes (Figure 11). A huge

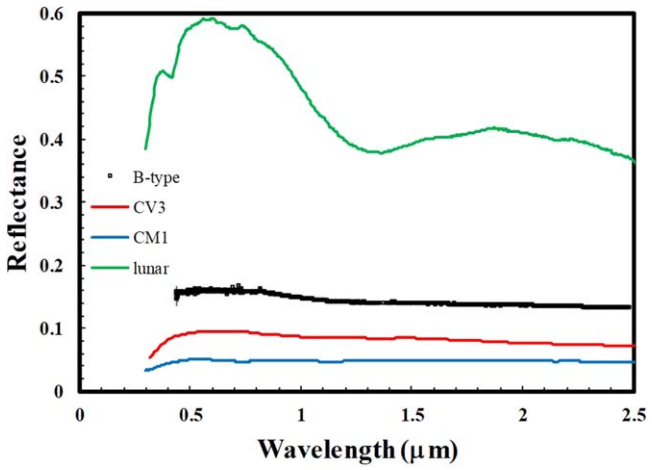


Figure 11. Reflectance spectra of objects classified as B-types. Plotted are (2) Pallas (DeMeo et al. 2009; black open squares), CV3 chondrite Allende (125–500 μm; red line), CM1 chondrite MET 01070 (<75 μm; blue line), and lunar meteorite (anorth) ALHA81005 (chip; green line). The Pallas spectrum was normalized to unity at 0.55 μm and then multiplied by 0.1587 (Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

range of meteorites have spectra that are classified as B-types (Table 3). Interestingly, a relatively large percentage of matches are with lunar meteorites, though these tend to be spectra of chips or slabs. The lunar spectra have blue spectral slopes but much stronger absorption bands. The lunar meteorite spectrum in Figure 11 is also much higher in albedo. We do not believe that a large percentage of B-types are lunar material; however, extremely space-weathered mafic material could be represented among B-types. A relatively large percentage (15%) of matches are also with aubrites. Aubrite surface compositions, which have high visual albedos (Ostrowski & Bryson 2019), for B-types could potentially be ruled out if the visual albedo of the asteroid is low.

Many of the other meteorite matches are also for slabs and chips. Approximately 32% of the matches are with CM or CV

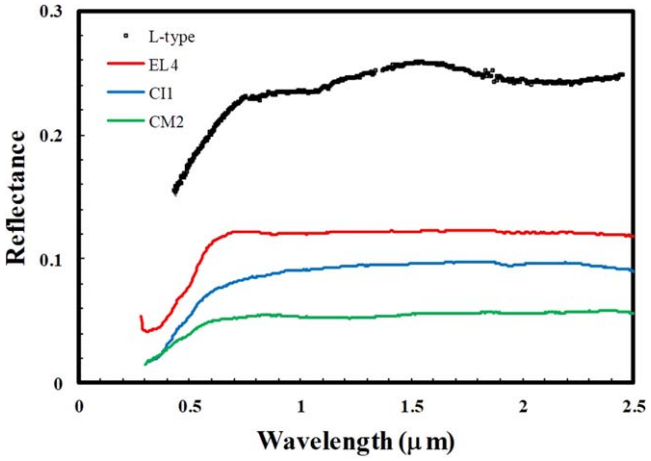


Figure 12. Reflectance spectra of objects classified as L-types. Plotted are (387) Aquitania (DeMeo et al. 2009; open black squares), EL4 chondrite MAC02747 (<45 μm; red line), CI1 chondrite (Orgueil) (<35 μm; blue line), and CM2 chondrite A-881280 (<125 μm; green line). The Aquitania spectrum was normalized to unity at 0.55 μm and then multiplied by 0.1900 (Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

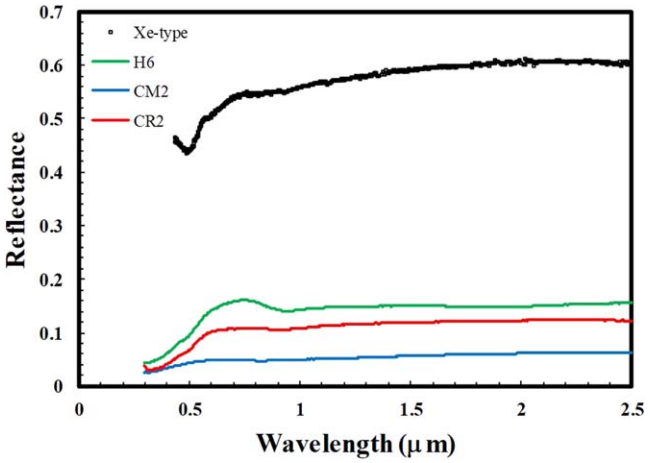


Figure 13. Reflectance spectra of objects classified as Xe-types. Plotted are (64) Angelina (DeMeo et al. 2009), CR2 chondrite EET 92042 (<125 μm; red line), CM2 chondrite LEW 88001 (<125 μm; blue line), and H6 chondrite Ozona (<50 μm; green line). The Angelina spectrum was normalized to unity at 0.55 μm and then multiplied by 0.474 (Shevchenko & Tedesco 2006). The error bars for the asteroid spectrum are 1σ.

material, which could have relatively featureless spectra and blue spectral slopes. CM and CV chondrites will tend to have blue spectral slopes for large grain sizes or flat surfaces.

4.2.11. L-types

L-types are typically linked with carbonaceous chondrite material (Figure 12) with enhanced abundances of calcium–aluminum inclusions (CAIs) (Burbine et al. 1992; Sunshine et al. 2008) due to the presence of a 2 μm absorption band stronger than the 1 μm band (characteristic of spinel). Spinel is a common constituent of CAIs. However, the proposed abundances of CAIs needed to produce such a feature are much higher than the abundances found in meteorites (Hezel & Russell 2008).

Many different meteorite types (e.g., enstatite chondrites, carbonaceous chondrites, ureilites, H chondrites) were given the L-type classification (Table 3). DeMeo et al. (2022) also previously noted the diversity of different spectral matches with L-type asteroids. Mahlke et al. (2023) found a number of spectral matches of L chondrites with CO and CV chondrites. A few CO and CV chondrites in the study were classified as L-types. However, CO and CV chondrite spectra typically do not have a stronger 2 μm band relative to the 1 μm band.

The primary issue in classifying these meteorites was that the primary choice presented by Bus–DeMeo was either L or Xe. Most meteorites did not have a strong absorption band shortward of ~0.55 μm that was similar to those found in Xe-types (Figure 13) such as (64) Angelina (Burbine et al. 2002). This absorption band has been attributed to oldhamite (CaS; Burbine et al. 2002), which is commonly found in aubrites (Watters & Prinz 1979). Angelina also has a much higher visual albedo than the plotted meteorites in Figure 13. This high visual albedo and the absence of low-Ca pyroxene absorption bands at ~0.9 and ~1.9 μm is also consistent with an aubritic composition. DeMeo et al. (2022) previously noted the rarity of meteorite spectra with spectral characteristics similar to Xe-types. We found that the L- and Xe-types were among the hardest classes to identify among the meteorite spectra.

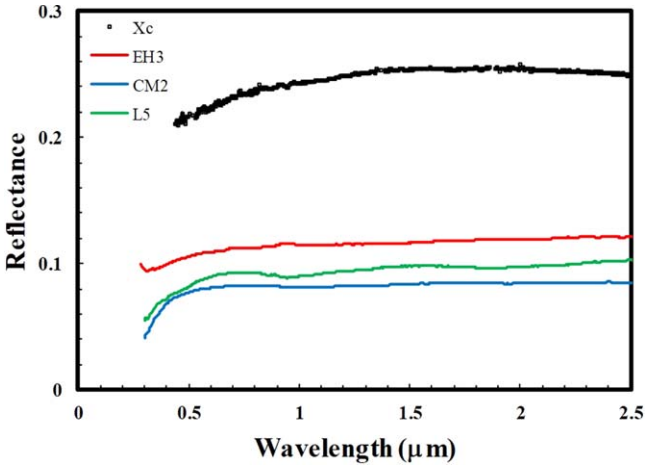


Figure 14. Reflectance spectra of objects classified as Xc-types. Plotted are (21) Lutetia (DeMeo et al. 2009; black open squares), EH3 chondrite KLE 98300 (<45 μm; red line), CM2 chondrite Dhofar 735 (<125 μm; blue line), and L5 chondrite Wittekrantz (25–250 μm, dark fraction; green line). The Lutetia spectrum was normalized to unity at 0.55 μm and then multiplied by 0.2212 (Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

4.2.12. Xc-types

Xc-types are typically linked with either carbonaceous chondritic material or iron meteorites because they have spectral properties intermediate between X- and C-types (Figure 14). Matches range from carbonaceous chondrites to enstatite chondrites to ordinary chondrites to achondrites (aubrites, ureilites; Table 3). Both Avdellidou et al. (2022) and DeMeo et al. (2022) previously noted a spectral similarity between enstatite chondrites and Xc-types. All of these meteorite groups can have relatively featureless visible and near-infrared reflectance spectra and moderate spectral slopes.

4.2.13. Other Classes

A few other classes, such as Xk, U, R, and Xe, had a substantial number of meteorite matches. Xe-types were

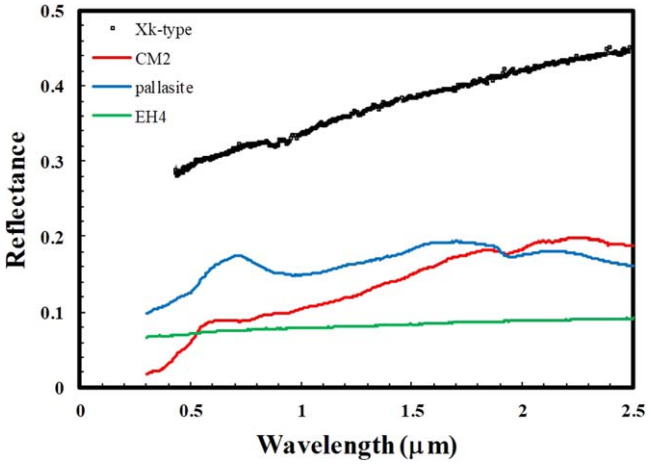


Figure 15. Reflectance spectra of objects classified as Xk-types. Plotted are (55) Pandora (DeMeo et al. 2009; black open squares), CM2 chondrite EET 83389 (<125 μm; red line), pallasite Y-8451 (slab; blue line), and EH4 chondrite Abee (<125 μm, heated at 800°C for 1 week in 10^{-5} atm of H_2; green line). The Pandora spectrum was normalized to unity at 0.55 μm and then multiplied by 0.3013 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

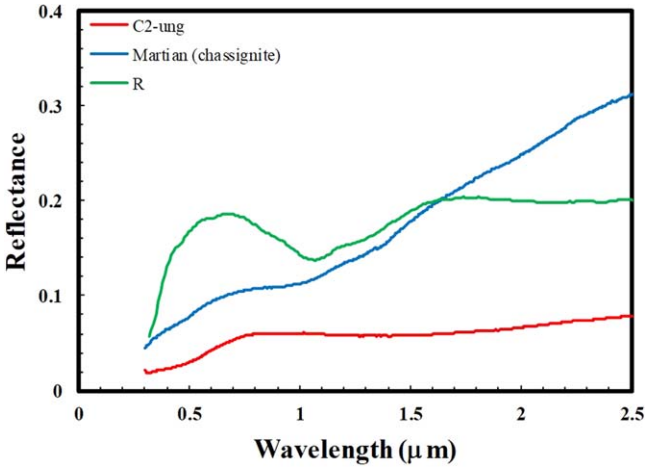


Figure 16. Reflectance spectra of objects classified as U-types. Plotted are ungrouped C2 chondrite Tagish Lake (<45 μm, heated at 200°C for 1 week with oxygen eater; red line), Martian meteorite (chassignite) NWA 2737 (<45 μm; blue line), and R chondrite Rumuruti (<150 μm; green line).

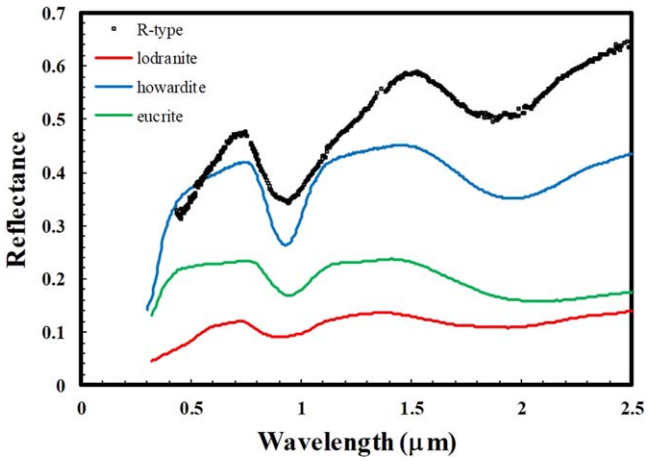


Figure 17. Reflectance spectra of objects classified as R-types. Plotted are (349) Dembowska (DeMeo et al. 2009), lodranite Lodran (chip; red line), howardite EET 87503 (<25 μm; blue line), and eucrite PCA 82501 (bulk; green line). The Dembowska spectrum was normalized to unity at 0.55 μm and then multiplied by 0.3840 (IRAS albedo; Tedesco et al. 2004). The error bars for the asteroid spectrum are 1σ.

previously discussed in the L-type section. Xk-types are typically linked with either CO/CV material or iron meteorites due to having spectral properties intermediate between X- and K-types (Figure 15). Many of the spectra that are classified as Xk (e.g., pallasites) are not spectrally similar to Xk asteroids but are worse matches for the other given choices such as Xc-, Xe-, C-, Ch-, or Xn-types.

U-types have reflectance spectra that are not consistent with any of the defined asteroid classes (Figure 16). Most of the U-type matches are with meteorites with distinctive absorption features that have shapes that are very different from those found in the defined Bus–DeMeo asteroid classes. As expected, a wide variety of meteorite types are classified as U-types. The difficulty in classifying the olivine-dominated R chondrite Rumuruti shows the problem that the Bus–DeMeo taxonomy has with classifying olivine-dominated reflectance spectra that are not red-sloped like A-types.

As mentioned earlier, R-types have spectral properties consistent with a low-Ca pyroxene–olivine assemblages (Figure 17) and have been typically linked with primitive achondrites (Abell 2003). The most-studied R-type asteroid is (349) Dembowska. Most R-type matches are with primitive achondrites (acapulcoites/lodranites) and HEDs. The small number of meteorite matches may be due to the lack of large numbers of primitive achondrite spectra in the RELAB spectral database. The HEDs that classify as R-types tend to have broader 1 μm band absorptions than typical HEDs. The numerous HED matches with R-types may be due to the large number of HEDs in the data set.

5. Conclusions

This study gives insight on the mineralogical diversity that may be found among the Bus–DeMeo classes. As expected from previous spectral studies, HEDs tend to dominate the V-type matches, while ordinary chondrites tend to dominate the S-complex and Q-type matches. CM chondrites represent a large proportion of the D- and X-type matches; however, there are also a substantial number of iron meteorites among the matches for these two classes. The large proportion of CM chondrite matches for the D- and X-classes is most likely due to the large number of CM chondrite spectra in our data set. Classes with weak-to-absent spectral features (e.g., C, B, L, Xc) contain a wide variety of meteorite assemblages with widely varying compositions and thermal histories. Visual albedos are vital for distinguishing between many carbonaceous chondritic and iron meteorite assemblages (e.g., DeMeo et al. 2022).

The Bus–DeMeo taxonomy is better at mineralogically classifying an object when the spectrum has "strong" absorption bands but does much worse with a spectrum that has "weak" to "absent" absorption features. The difficulty of the Bus–DeMeo taxonomy in accurately classifying many meteorite spectra and the rarity of many meteorite matches among the asteroid classes implies that the reflectance spectra of observed asteroids do not fully cover the spectral range found in meteorites. Many biases do exist in our meteorite spectral data set since the number of spectra of a particular meteorite group is a function of many factors, such as scientific interest (e.g., linkage to a spacecraft mission target) and

abundance in our meteorite collections. We would not expect the Bus–DeMeo taxonomy to classify lunar and Martian meteorite spectra particularly well mineralogically because the Bus–DeMeo taxonomy was defined using asteroid spectra.

However, a number of nonlunar and non-Martian meteorite types are also difficult to classify. For example, olivine-dominated meteorites that do not have red-sloped spectra do not have a corresponding asteroid class. Meteorites may not be sampling the "space-weathered" surfaces of these olivine-dominated bodies. Meteorites with reflectance spectra similar to many "rarer" asteroid classes (e.g., A, Xn, T) appear extremely uncommon. Biases in being transported to Earth (e.g., strength) or making it through the atmosphere will affect the probability that a particular type of meteorite arrives on the Earth's surface. Some types of meteoritic material may also be currently rare in the asteroid belt or originate from bodies that are extremely small and difficult to observe spectroscopically. We suggest that the Bus–DeMeo taxonomy give an option for classifying a spectrum as U when none of the given choices match the spectral characteristics of the spectrum.

No asteroid class perfectly groups meteorites together with similar mineralogies and thermal histories. Detailed analyses of absorption bands are necessary for interpreting the mineralogy of most objects, which might be crucial for mitigation efforts for an Earth-impacting object. Only more sample-return missions will give the necessary insights into the diversity of each asteroid class.

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Appendix

List of meteorite spectra used in the study (Table A1). Listed are the meteorite name, its classification (Meteoritical Bulletin Database 2024), the asteroid class given for its spectrum, its RELAB spectrum ID, its particle size or chip/slab, and any note on the characteristics of the sample. The terms chip and slab are used rather interchangeably. We note that the Milliken (2020) spectral database, where information on the spectra was obtained, sometimes had conflicting information.

Table A1
List of Meteorite Spectra Used in This Study with Their Classifications, the Asteroid Class Given for This Spectrum, Its Particle Size, and Any Note on the Sample

Meteorite Name	Classification	Asteroid Class	RELAB Spectrum ID	Particle Size	Note
A-87272	Eucrite-mmict	V	camp94	<25 μm	⋯
A-881280	CM2	L	c1mp56	<125 μm	⋯
A-881334	CM2	Xc	c1mp35	<125 μm	⋯
A-881394	Eucrite-cm	V	c1mp136a	500–2000 μm	⋯
A-881458	CM2	K	c1mp36	<125 μm	⋯
A-881526	Diogenite	V	camp95	<25 μm	⋯
A-881594	CM2	B	c1mp57	<125 μm	⋯
A-881595	CR2	Xe	c1mp39	<125 μm	⋯
A-881632	CO3	L	c2mp104	<125 μm	⋯
A-881655	CM2	K	c1mp55	<125 μm	⋯

(This table is available in its entirety in machine-readable form in the online article.)

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